QUENTIN **BONASSIES**

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I have defended my PhD on April 21st, 2026, at CERFACS. The PhD was co-supervised between CERFACS, CNES and CLS and was focused on data assimilation using remote sensing data (SWOT, Sentinel-1 & -2) for flood hydrodynamic modeling.

EXPERIENCES

OCTOBER 2022 — APRIL 2026

PHD (CNES PHD — COFUNDED BY CNES AND CLS)

CERFACS, TOULOUSE (FRANCE)

Doctorate co-supervised by L. Cassan (CERFACS, cassan@cerfacs.fr), S. Peña-Luque (CNES, santiago.penaluque@cnes.fr), C. Fatras (CLS, cfatras@groupcls.com) and S. Ricci (CERFACS, ricci@cerfacs.fr).

The thesis manuscript is entitled: "Data assimilation of remote sensing flood extent for flood forecasting: Benefits of data provided by the SWOT satellite."

The first part of the PhD was focused on the implementation of the Chan-Vese metric in a sequential data assimilation algorithm (EnKF and ETKF) in Telemac-2D. This was followed by a two-month internship at the Jet Propulsion Laboratory (see below) to set up a chain between a hydrological model and a hydrodynamic model with data assimilation. Then the thesis work continued with the assimilation of the Surface Water and Ocean Topography (SWOT) satellite data using L2 Nodes products in the hydraulic model Telemac-2D. It continues with a study to understand the L2 Pixel Cloud SWOT data for flood extent extraction. Then, my work focused on developing a Telemac-2D model near Chinon (France) and the assimilation of SWOT-derived flood extent using the Chan-Vese assimilation method developed during the thesis in a real flood event.

FEBUARY 2024 -MARCH 2024

DOCTORAT INTERNSHIP

JET PROPULSION LABORATORY, PASADENA, CALIFORNIA (USA)

Internship co-supervised by C.H. David (JPL, cedric.david@jpl.nasa.gov), H. Oubanas (INRAE UMR G-EAU, hind.oubanas@inrae.fr), S. Ricci (CERFACS), and A. Piacentini (CERFACS, piacentini.palm@gmail.com).

The internship consists of modeling water discharges in the Garonne watershed using the hydrological RAPID model. The internship is part of the SCO-FloodDAM project (CNES) and AIST-IDEAS (JPL) for the creation of digital twins for flood forecasting. Modeling includes setting up data assimilation (Kalman Filter) experiments using data from the Vigicrue network. The addition of SWOT L4 Discharge data into the assimilation framework was explored during the PhD after the internship.

OCTOBER 2021 – SEPTEMBER 2022

HYDRAULIC MODELING ENGINEER

ARTELIA, ÉCHIROLLES (FRANCE)

In the numerical modeling team supervised by O. Bertrand (Olivier.BERTRAND@arteliagroup.com)

2D and 3D modeling of river and marine hydraulics using the OpenTelemac suite. Consideration of thermal, meteorological and/or ecological effects. Feasibility and/or impact studies in 2D/3D

hydraulic modeling with Telemac software: sediment discharge problems at sea, studies linked to hydrocarbon accidents at sea or in estuaries, cold water recirculation in a port, analysis of a thermal basin before discharge into a river. Use of an HPC server, creation of unstructured triangular meshes, writing of Fortran scripts, calibration/validation of models, analysis of results (Python).

MARCH 2021 - SEPTEMBER 2021

FINAL YEAR INTERNSHIP

ARTELIA, ÉCHIROLLES (FRANCE)

In the numerical modeling team co-supervised by M-P Daou and M. Gonzales de Linares.
Rewriting and developing in Python a 1D fluvial sediment transport modeling software (hydraulic - solid transport coupling) internal to Artelia: simulation of the evolution of river bottoms and solid transport during floods or hydrological cycles. Conversion from Visual Basic 6 to Python 3, implementation of a graphical user interface (Qt), database management (sqlite3), software development (Python), addition of functionalities to existing software, hydraulic and solid transport modeling.

JUNE 2020 – SEPTEMBER 2024

RESEARCH INTERNSHIP

OMP - LEGOS, TOULOUSE (FRANCE)

Supervised by C. Ulses.
Study of a regional 3D coupled physical/biogeochemical model of the Mediterranean Sea (SYMPHONIE/Eco3M-S; Estournel et al., 2016; Ulses et al., 2016; Kessouri et al., 2017; 2018). Verification of model outputs related to the carbonate system. Graphical and statistical comparison of model output within situ observational data. Study of the formation of Levantine Intermediate waters by studying the carbonate system in the Levantine Basin in the eastern Mediterranean. High level of autonomy in response to the specific conditions associated with COVID-19.

PUBLICATIONS

- **Q. Bonassies**, C. Fatras, S. Pena Luque, P. Dubois, A. Piacentini, L. Cassan, S. Ricci, and T. H. Nguyen, “A Comprehensive Study of Surface Water and Ocean Topography Pixel Cloud Data for Flood Extent Extraction”, Remote Sens. Env., 333, 115101, 2026.
  DOI: 10.1016/j.rse.2025.115101
- **Q. Bonassies**, T. H. Nguyen, L. Cassan, A. Piacentini, S. Ricci, C. Emery, C. Fatras, S. Pena Luque, and R. Rodriguez Suquet, “Assimilation of SWOT Altimetry Data for Riverine Flood Reanalysis: From Synthetic to Real Data”, IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens. (accepted).
  DOI: 10.1109/JSTARS.2026.3659808
- T. H. Nguyen, A. Piacentini, S. Ricci, L. Cassan, S. Munier, **Q. Bonassies**, and R. Rodriguez Suquet, “Remote Sensing Data Assimilation with a Chained Hydrologic-hydraulic Model for Flood Forecasting”, Water Resour. Res., 62, e2024WR039468, 2026.
  DOI: 10.1029/2024WR039468
- M. Tom, C. H. David, K. M. Marlis, P. A. Zimdars, **Q. Bonassies**, J. Wade, A. Cerbelaud, M. Bonnema, T. Pavelsky and T. Huang, “Event-based serverless environmental modeling on the cloud: A pedagogical guide and river case study”, Environmental Modelling & Software, 106753, 2026.
  DOI: 10.1016/j.envsoft.2025.106753

CONFERENCE PRESENTATIONS, POSTERS & OUTREACHS

Telemac User Conference 2023 — Oral — Bonassies *et al.*, *Front type implementation for flood extent and flood modelling image assimilation*

Colloque de la Société Hydrotechnique de France 2023 — Poster — Bonassies *et al.*, *SAR data assimilation as front-like data for flood prediction*

SWOT Science Team Meeting 2025 — Oral — Bonassies *et al.*, *A comprehensive study of Surface Water and Ocean Topography (SWOT) Pixel Cloud data for flood extent extraction*

SWOT Science Team Meeting 2025 — Poster — Nguyen, Bonassies *et al.*, *Assimilation of SWOT Nodes products into basin-scale hydraulic models*

Youtube video — *made with the Office for Climate Education* *| Comment SWOT améliore les prévisions des inondations ? | Quentin Bonassies (Link to the video)*

EDUCATION

2018 – 2021

ENGINEERING DEGREE

INP-AGROTOULOUSE (OLD NAME: INP-ENSAT)

- Soil Science and Agronomy Engineering school.
- Specialization in science, ecology, and water modeling (Specialization perform within the three INP school of Toulouse — INP-ENSIACET, INP-ENSEEIHT and INP-ENSAT).

2016 – 2018

PREPARATORY CLASSES FOR THE "GRANDES ÉCOLES" — BCPST

LYCÉE HOCHE, VERSAILLES

2016

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TOOLS AND SKILLS

Python, Fortran, Shell
HPC, Git, Docker
GDAL, QGIS
OpenTelemac, RAPID model

Matlab, R, VBA, Cython, C++
ArcGIS, Qt, SQLite3

English (C1 - TOEFL ITP (2020): 640/677)
Good scientific writing and reading

Remote Sensing — SAR; large swath interferometer (SWOT); Optical
Data assimilation (mostly Kalman Filtering algorithms)
Hydraulic and hydrological modeling

ACTIVITIES AND CENTRES OF INTEREST

Photography
Run, Bike, Badminton, Orienteering
Hiking

Nature
Aquarium enthusiasts
Saxophone

OTHER EXPERIENCES

Stays in the USA in total immersion within American families (during high school and preparatory classes — 2 to 5 weeks every year for 4 or 5 years)